

How did they generate meal descriptions?

First, they took **real meal data** from two source datasets: **What We Eat in America (WWEIA)** and **FAO/WHO GIFT**. They processed the meals, attached nutrition labels, and for the FAO/WHO GIFT meals they also converted many gram amounts into **everyday serving sizes** like cups or tablespoons.

Then they used **GPT-4o-mini** to turn those structured food records into **natural-language meal descriptions**. For each food item or meal, they prompted GPT-4o-mini to generate **five different descriptions**, and then they **randomly chose one** as the final query.

After that, they did **human verification**. One of the authors checked whether the generated descriptions correctly included the right **food names** and **serving sizes**, because GPT-4o-mini sometimes left out an item or a serving amount.

They used WWEIA (US data) and FAO/WHO GIFT (global data) as sources. Why?

My understanding from paper is:

1. WWEIA = U.S. data

This is a U.S. food survey where trained interviewers recorded what people ate and how much they ate over 24 hours. The authors used it to get real meal items and nutrition labels from the United States.

2. FAO/WHO GIFT = global data

This is an international food-consumption database. The authors used it to collect meals from 10 other countries so the dataset would not be only American food.

So together:

- WWEIA gave them U.S. meals
- FAO/WHO GIFT gave them non-U.S. meals from around the world
- then they turned those real meal records into natural-language descriptions for NutriBench

Using both sources made the dataset more diverse, because it includes meals from 11 countries, not just one food culture. That helps test whether AI can estimate nutrition for different kinds of foods and eating habits.

Why they used GPT-4o-mini to generate natural language descriptions from raw food records?

They had raw food records, which are basically structured database entries, not normal sentences.

A raw record might look like: food name, serving size, and eating occasion — for example, something like “banana, 1 cup, breakfast” or a list of foods with amounts.

Then they asked GPT-4o-mini to rewrite those records into normal human-sounding meal descriptions.

So instead of database-style text, the model would generate something like: “For breakfast, I had a cup of oatmeal with half a banana and a glass of orange juice.” The paper says they asked GPT-4o-mini to create five different natural-language versions for each input, then randomly picked one.

After that, a human checked the output to make sure GPT-4o-mini had not forgotten any food item or serving size.

This was important because the model sometimes left out foods or portions, especially in more complex meals.

Btw Total meal descriptions: 11,857.

The dataset contains these 7 columns:

Column	What it contains
<code>meal_description</code>	A sentence describing the meal in natural language
<code>carb</code>	Carbohydrate content (in grams)
<code>fat</code>	Fat content (in grams)

energy	Energy/calories (in kilocalories)
protein	Protein content (in grams)
country	Country code (e.g. "ZMB" = Zambia, "USA" = United States)
serving_type	Either <code>metric</code> or <code>natural</code> (how servings are measured)

If I wanted to add Bangladesh, what data source would I need?
I think **BIRDEM Food Table**.

FCTB too:

https://www.researchgate.net/publication/255901003_final_report_of_Food_Composition_Table_for_Bangladesh

<https://pubmed.ncbi.nlm.nih.gov/35763921/>